

Theory of Magnetism

The Magnetism explains how certain materials can **attract or repel each other without touching**.

❖ Basic Idea

Magnetism is caused by **moving electric charges inside atoms**. Every material is made of tiny particles called atoms, and inside them, electrons move. This movement creates tiny magnetic effects.

- In most materials → these tiny magnetic effects cancel out
 - In magnetic materials (like iron) → they **line up in the same direction**, creating a strong magnet
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❖ Magnetic Domains

Inside magnetic materials, there are small regions called **domains**:

- Each domain acts like a tiny magnet
- In an unmagnetized object → domains are arranged randomly
- In a magnet → domains are aligned in one direction

☞ When domains align, the material becomes a magnet.

❖ Magnetic Poles

Every magnet has two ends called poles:

- **North pole (N)**
- **South pole (S)**

Important rules:

- Like poles repel (N–N, S–S)
- Unlike poles attract (N–S)

You cannot get a magnet with only one pole—if you cut a magnet, each piece will still have both poles.

❖ Magnetic Field

A magnet creates an invisible region around it called a **magnetic field**.

- This is where magnetic force can be felt
 - The force is strongest near the poles
 - It acts without physical contact
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❖ Earth as a Magnet

The Earth's magnetic field makes our planet behave like a giant magnet.

- This is why a compass always points North-South
 - It helps in navigation
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❖ Temporary vs Permanent Magnetism

- **Temporary magnets:** Lose magnetism quickly (e.g., soft iron)
 - **Permanent magnets:** Keep magnetism for a long time (e.g., steel)
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In Simple Words

Magnetism happens when tiny magnetic parts inside a material **work together in the same direction**, creating a force that can **pull or push objects without touching them**.